Historical Passenger Survival & Risk-Factor Analysis

Excel Capstone | RMS Titanic supplied sample

Prepared by: Chima Praise Chizzy

Decision question: What patterns appear in the supplied passenger records, and what must be validated before using them for decisions?

Scope: 418 records from the supplied Titanic.csv — not the full historical manifest.

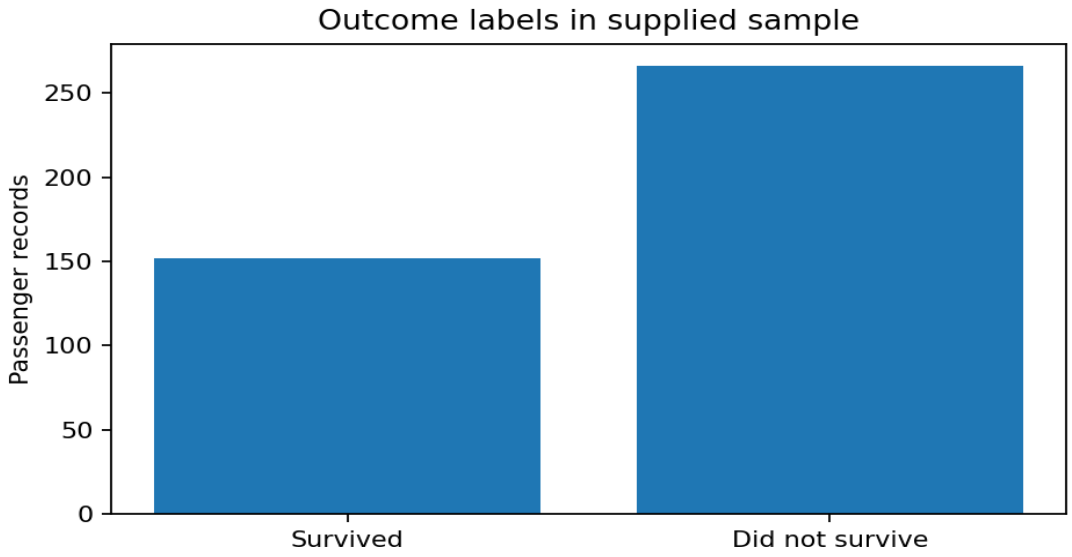
Bottom line: validate the outcome labels before drawing conclusions

Executive summary

• 152 of 418 records are labelled as survived (36.4%).

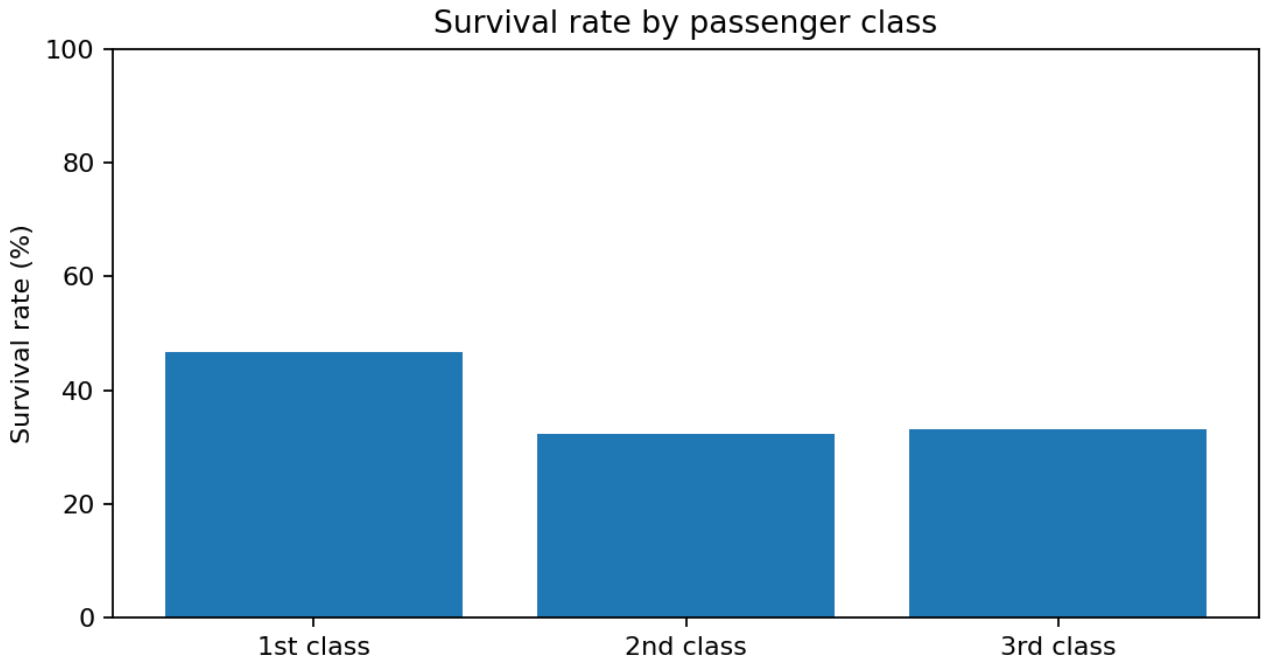
• Every female record is labelled survived; every male record is labelled not survived. This perfect separation is the primary data-quality warning.

• First class has a higher labelled survival rate than second or third class, but the observed patterns are descriptive and not causal.



Class differences exist, but the sample needs careful interpretation

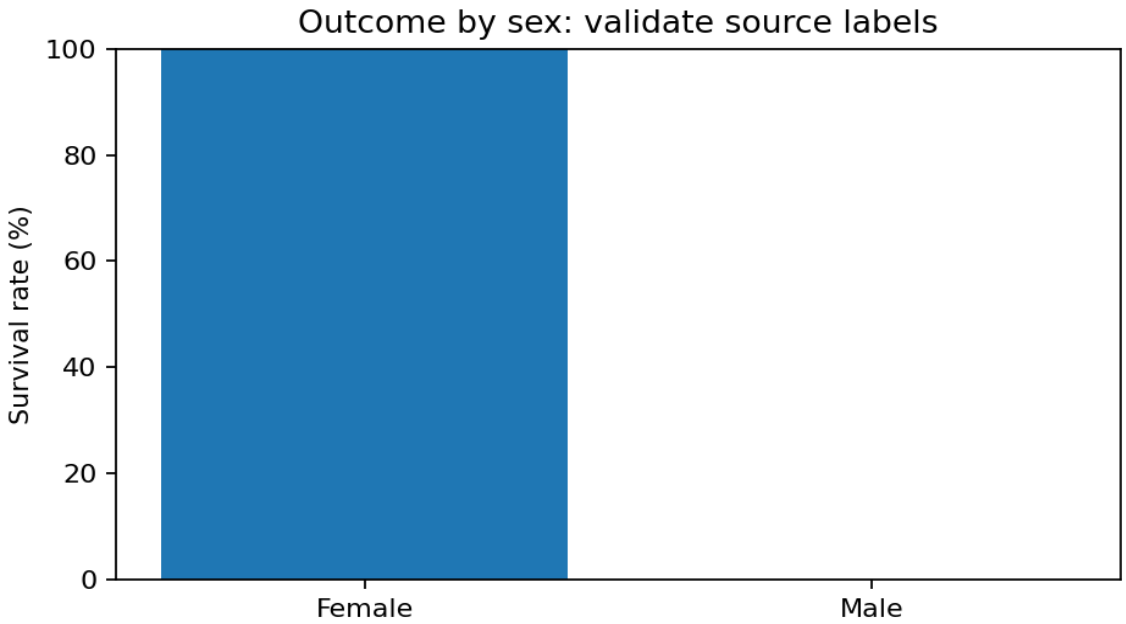
Finding 1



First class: 46.7% | Second class: 32.3% | Third class: 33.0%

The sex-outcome pattern is too perfect to ignore

Finding 2 — data validation



- Female: 152/152 labelled survived. Male: 0/266 labelled survived.
- Confirm the meaning and provenance of Survived with the instructor/source owner before using it as a risk predictor.

Fare and family-size relationships are weak and potentially confounded

Finding 3

- Fare and survival label correlation: r ≈ 0.192 (weak positive association).
- Family size and survival label correlation: r ≈ 0.162 (weak positive association).
- Mean analysis fare falls from about 94.28 in first class to 22.20 in second and 12.46 in third class.
- Small sample sizes in some age and family-size groups make rates unstable.

Recommended next steps

Decision and limitations

• Validate outcome labels and confirm the intended scope with the instructor.

• Finish Power Query staging/reference queries, documented transformations and row reconciliation in Excel.

• Build at least two PivotTables and PivotCharts, connect Slicers, and add a GETPIVOTDATA KPI card.

• Retain outliers with documented flags; show counts beside rates; avoid causal language.

• Any modern passenger-safety model requires representative data, independent validation and additional operational context.